

SCHOOL OF COMPUTING (SEM II – 2024/2025)
SECI2143 – PSDA

Exercise (Chapter 5 – Part3)

QUESTION 1

Company XYZ distributes chocolate drink known as Micho in bottles labelled 32 ml. Company ABC, an auditor company, randomly selects 50 of these bottles, measures their content and obtains a sample mean of 31.8 ml and a sample standard deviation of 0.75 ml. Using a 0.01 significance level, test the auditor company claims that the Company XYZ is cheating consumers. The Company XYZ is cheating consumers if it distributions bottles containing less than 32 ml of Micho.

- a. State the null and alternative hypotheses. (2 marks)
- b. Determine and calculate the appropriate test statistic. (2 marks)
- c. Find the critical value. (2 marks)
- d. Draw and shade the critical region (rejection area) and state the critical value in the distribution. (2 marks)
- e. Is it true that the Company XYZ is cheating consumer? (1 mark)

QUESTION 2

James and his sister, Alison, each deliver 30 newspapers and each are paid the same amount. Alison claims that the equal payment is unfair as she takes on average longer to distribute the newspaper compare to her brother, James. To test her claim, her father records their delivery times (in minutes) as shown below.

Table 1: Delivery times (minutes)

James	Alison
45	49
30	42
39	39
32	45
34	38
43	49
38	43
35	36
43	33
39	41
32	36
34	-
$\bar{x}_1 = 30$ $s_1 = 4.92$	$\bar{x}_1 = 33$ $s_1 = 5.25$

- What is the assumption required for this hypothesis analysis? (1 mark)
- State the null and alternative hypotheses. (2 marks)
- Variance of both population are unknown and assume both are similar, test whether Alison's claim is justified, using a 5% level of significance. (7 marks)
- Based on result in (c), what change would you make to your test procedure so that the conclusion of the hypothesis is different than (c)? (2 marks)

QUESTION 3

The following F -test was generated for the data set that contains 37 ceramic strength measurements for two batches of material. The summary statistics for each batch are shown below.

BATCH 1:

NUMBER OF OBSERVATIONS	=	16
MEAN	=	688.9987
STANDARD DEVIATION	=	65.54909

BATCH 2:

NUMBER OF OBSERVATIONS	=	21
MEAN	=	611.1559
STANDARD DEVIATION	=	61.85425

Use 95% significance level to test whether the standard deviation of batch 1 is greater than the standard deviation of batch 2. (8 marks)

QUESTION 4

A survey was conducted to investigate the age distribution of persons arrested for careless driving. They are distributed in a five different age groups. To test the hypothesis that number of arrested drivers are equal among these five groups, a random sample of 100 persons arrested for careless driving is identified. Given the hypotheses as below,

$$H_0: p_1 = p_2 = p_3 = p_4 = p_5$$

$$H_1: \text{At least one of the } p_i \text{ is not equal}$$

- If the resulting of χ^2 value were 16.5, what conclusion would you reach when using a test with significance level 0.01? (2 marks)
- What conclusion would be appropriate at significance level 0.05 if the resulting of χ^2 value were 8.2? (2 marks)
- If there were only four groups of age rather than five groups, what would you conclude if χ^2 value were 9.5 and a test with $\alpha = 0.025$ was used? (2 marks)

QUESTION 5

As a part of an industrial training program, some trainees are instructed by Method A, which is straight teaching-machine instruction, and some are instructed by Method B, which also involves the personal attention of an instructor. Random samples of size 10 are taken from large groups of trainees instructed by each of these two methods, and the scores they obtained in an appropriate achievement test are shown in Table 1.

Table 1: Scores of trainees

Method A	71	75	65	69	73	66	68	71	74	68
Method B	72	77	84	78	69	70	77	73	65	75

Given: $s_A^2 = 11.3333$, $s_B^2 = 29.1111$

Use 0.05 level of significance to test the claim that Method B is more effective. Assume that the populations sampled can be approximated closely with Normal distribution having the same variance.

QUESTION 6

In a study of the effectiveness of physical exercise in weight reduction, a group of 12 persons participate in a prescribed program of physical exercise for one month showed the following results (Table 2):

Table 2: Weight reduction of participants

Weight before (kg)	95	81	76	96	82	87	71	82	77	83	81	65
Weight after (kg)	89	77	77	94	80	86	72	82	74	81	78	63

Use the 0.01 level of significance to test whether the prescribed program of exercise is effective.